

tasks $x \in (\alpha, \hat{x})$, augmented productivity is $\tilde{\gamma}_J(x) = (1 + \delta_J)\gamma_J(x)$, so the ratio of post- to pre-AI labor demand is

$$\frac{l_J^{*\text{post}}(x)}{l_J^{*\text{pre}}(x)} = \left(\frac{P^{\text{post}}}{P^{\text{pre}}} \right)^\sigma (1 + \delta_J)^{\sigma-1}. \quad (31)$$

Since GenAI weakly reduces unit costs on every task—automated tasks satisfy $c_A(x) < w_J/\gamma_J(x)$ by assumption, augmented tasks have unit cost $w_k/((1 + \delta_k)\gamma_k(x)) \leq w_k/\gamma_k(x)$, and the threshold reallocation only occurs when it is cost-reducing—the post-AI cost index satisfies $P^{\text{post}} \leq P^{\text{pre}}$, with strict inequality whenever $\alpha > 0$ or $\delta_k > 0$ for some k .³⁵ Write $\rho \equiv (P^{\text{post}}/P^{\text{pre}})^\sigma \in (0, 1]$.

The ratio (31) is uniform across common junior tasks, so the sign of the productivity effect is determined by whether $\rho \cdot (1 + \delta_J)^{\sigma-1}$ exceeds, equals, or falls below one:

- $\sigma < 1$: $(1 + \delta_J)^{\sigma-1} \leq 1$ (with equality iff $\delta_J = 0$) and $\rho \leq 1$. The product is ≤ 1 , with strict inequality whenever $\delta_J > 0$ or $\alpha > 0$. The productivity effect is non-positive.
- $\sigma > 1$: $(1 + \delta_J)^{\sigma-1} \geq 1$ but $\rho \leq 1$. Within-task augmentation raises labor demand per task, while the reduction in the cost index lowers it. The sign is ambiguous in general.
- $\sigma = 1$: $(1 + \delta_J)^0 = 1$, so the ratio reduces to $\rho = P^{\text{post}}/P^{\text{pre}} \leq 1$. The within-task augmentation effect vanishes and only the cost-index effect remains. The productivity effect is non-positive, and strictly negative whenever $P^{\text{post}} < P^{\text{pre}}$.³⁶

Sufficient condition for net decline when $\sigma \leq 1$.

When $\sigma \leq 1$, the productivity effect is non-positive, so

$$\Delta L_J^* \leq - \int_0^\alpha l_J^{*\text{pre}}(x) dx + \int_{\hat{x}}^{\hat{x}'} l_J^{*\text{post}}(x) dx. \quad (32)$$

³⁵The CES cost index P is increasing in each task's unit cost. Since cost minimization assigns each task to the cheapest available input, and GenAI either automates tasks at lower cost or augments human productivity, every task's effective unit cost weakly decreases after GenAI adoption.

³⁶With $\sigma = 1$, the production function is $Y = \exp\left(\int_0^1 \ln y(x) dx\right)$ and cost minimization equalizes expenditure across tasks: $p(x)y(x) = P \cdot Y$ for all x . Since $p(x) = w_J/\gamma_J(x)$ and $y(x) = \gamma_J(x)l_J(x)$, we obtain $w_J l_J^*(x) = P \cdot Y$, i.e., $l_J^*(x) = P Y/w_J$. After GenAI adoption, $l_J^{*\text{post}}(x) = P^{\text{post}} Y/w_J < P^{\text{pre}} Y/w_J = l_J^{*\text{pre}}(x)$ whenever $P^{\text{post}} < P^{\text{pre}}$.

The reallocation effect is bounded above by $(\hat{x}' - \hat{x}) \cdot \bar{l}$, where $\bar{l} = \sup_{x \in [\alpha, \hat{x}']} l_J^{*\text{post}}(x)$. The displacement effect $\int_0^\alpha l_J^{*\text{pre}}(x) dx > 0$ for any $\alpha > 0$. Therefore $\Delta L_J^* < 0$ whenever

$$(\hat{x}' - \hat{x}) \cdot \bar{l} < \int_0^\alpha l_J^{*\text{pre}}(x) dx. \quad (33)$$

Note that (33) is only a sufficient condition for a net decline in junior labor demand, not a characterization of all parameter configurations under which the decline occurs; the net effect can be negative even when this bound is not satisfied, because the bound replaces the productivity effect with zero.

When $\delta_J = \delta_S$, the augmented productivity ratio $\tilde{\gamma}_S(x)/\tilde{\gamma}_J(x) = \gamma_S(x)/\gamma_J(x)$ is unchanged, so $\hat{x}' = \hat{x}$ exactly. The reallocation effect is zero and $\Delta L_J^* < 0$ follows immediately. More generally, by the implicit function theorem (applicable since $\frac{d}{dx} \ln(\gamma_S/\gamma_J)|_{\hat{x}} > 0$ by Assumption 1), $\hat{x}'$ is continuous in (δ_J, δ_S) . Since $\hat{x}' - \hat{x}$ vanishes at $\delta_J = \delta_S$ while the displacement effect is bounded away from zero, there exists a neighborhood around $\delta_J = \delta_S$ in which (33) holds.

A.7 AI Usage Patterns Across the Seniority Spectrum

This appendix provides external descriptive evidence on how GenAI is used for tasks at different levels of the occupational seniority hierarchy, using data on GenAI usage patterns that are independent of the firm-level employment data in the main analysis. The analysis draws on the Anthropic Economic Index (AEI), a public dataset that classifies real-world interactions with a large language model by the occupational tasks they serve and the nature of human–AI collaboration. We merge these GenAI usage measures with occupation-level seniority indicators from Revelio Labs/LinkedIn to document systematic differences in the *mode* of GenAI engagement across junior-type and senior-type tasks. The goal is not to establish causal effects but to provide theory-motivated descriptive evidence that complements the conceptual framework in Section 3: the framework predicts that junior-type tasks are disproportionately exposed to automation, while senior-type tasks—conditional on remaining human-performed—are more likely to involve augmentation-type GenAI usage. Note that while the main analysis studies within-firm junior versus senior employment, Appendix A.3 uses occupation-level seniority as a proxy for the type of tasks typically performed by more junior or more senior workers—it captures cross-occupation variation in task complexity, not the within-firm hierarchy itself. The data allow us to examine whether observed GenAI usage patterns are consistent with the framework’s prediction.

A.7.1 Data Sources

Anthropic Economic Index (AEI). The AEI dataset (Appel et al., 2026) is constructed from a large sample of anonymized conversations with Claude, Anthropic’s large language model, during the week of November 13–20, 2025. Each conversation is algorithmically matched to the O*NET task it most closely corresponds to and classified into one of five *collaboration modes* that characterize how the user interacts with the AI:

- **Directive:** the user instructs the GenAI to produce an output with minimal back-and-forth. The GenAI executes a well-specified request—analogueous to delegating a task to a subordinate.
- **Learning:** the user seeks to acquire knowledge or understanding from the AI, using

it as an informational resource or tutor.

- **Task iteration:** the user and GenAI collaboratively refine an output through repeated exchange, with the user providing feedback and the GenAI revising its work.
- **Validation:** the user presents work or reasoning to the GenAI for checking, verification, or quality assurance.
- **Feedback loop:** a hybrid mode combining elements of directive and iterative interaction.

We use these collaboration modes as an interpretive proxy for the framework’s conceptual distinction between displacement and augmentation, recognizing that this mapping is a classification choice rather than a direct measurement of economic substitution. Specifically, we classify directive use and feedback loops as *displacement-type* interactions, in which GenAI usage tilts toward substitution for human task execution, and learning, task iteration, and validation as *augmentation-type* interactions, in which GenAI complements human judgment and expertise. We use data from both the Claude.ai consumer platform and the first-party API, though we focus primarily on Claude.ai usage in the reported results.

Revelio Labs/LinkedIn workforce composition. Our primary seniority measure is based on Revelio Labs data derived from public LinkedIn profiles. For each occupation, we compute the average seniority level of active workers as of January 2023 (pre-ChatGPT) using Revelio’s 1–7 seniority scale, and standardize to a z-score. The Revelio measure captures the *actual* seniority composition of the workforce—whether an occupation is predominantly staffed by junior or senior workers—which directly corresponds to the junior–senior distinction emphasized in the conceptual framework. We restrict the sample to occupations with at least 50 observed workers and partition occupations into terciles based on this measure (“low/junior-type,” “medium,” “high/senior-type”).

O*NET experience requirements. As a complementary validation, we also report the average years of related work experience required for each occupation from the O*NET Education, Training, and Experience database (version 30.0). This normative measure